

0:00	Title	<i>“Good morning. My name is Martin Maxa, from the Department of Measurement – here at this faculty. I worked on this together with Jakub Svatoš, from the same department.”</i> ■ subject: detection, classification, localization of impulsive sounds – gunshots as the target class ■ core of the contribution: comparison of cepstral extractors + two classifiers; localization = second function of the same network
0:45	Motivation	<i>“This work is part of the development of an acoustic sensor unit for the protection of soft targets.”</i> ■ acoustic channel complements cameras: omnidirectional, no line of sight ■ [CUT] three functions: detection, classification (to calibre), localization – in noisy, reverberant outdoor environments ■ three questions: (1) which cepstral representation separates gunshots from similar false alarms, (2) NN or SVM on those features, (3) frame length vs. reflections
1:50	Signature	<i>“Both detection and localization depend on what the signal actually looks like, so let us start there. A gunshot recording contains one or two components, depending on the speed of the projectile.”</i> ■ subsonic → muzzle blast alone; supersonic → shock wave ahead of it <i>“the recording shown is a supersonic shot, so both are present”</i> ■ shock wave ≈ 2.5 ms, <i>supersonic only</i> – [CUT] Mach cone along the trajectory, N-shaped pressure profile ■ [CLICK] muzzle blast ≈ 45 ms later, <i>present for every shot</i> – propellant gases at the barrel, speed of sound, 1–3 ms ■ between and after = reflections, not source signature – <i>“relevant to frame length; I return to it in the results”</i>
3:00	Single-domain	<i>“Neither signal domain is sufficient on its own.”</i> ■ spectral analysis of a frame → characterises the environment, not the source ■ time-domain pattern → nonlinear dispersion, obstacles in the path ■ ⇒ combine both domains – which is what a cepstral representation does
3:30	System	<i>“The system architecture is a distributed network of stand-alone sensor units and a remote server.”</i> ■ unit monitors continuously → on an impulsive event, transmits the segment ■ server: classification + 2D position from several units ■ several units: coverage – and a 2D position needs <i>at least two bearings</i>
4:20	Localize	<i>“Localization. Each sensor unit carries two MEMS microphones, almost twenty centimetres apart, sampled at 44.1 kilohertz.”</i> ■ TDoA between channels, at most ≈ 24 samples for this geometry ■ $\Theta = \arcsin(c \Delta n / (d f_s))$ – speed of sound $\times$ delay, over spacing $\times$ sampling rate ■ [CLICK] one unit = a direction; two bearings intersect = position estimate ■ evaluated on 70 events, 9 angles, -45° to $+60^\circ$ – cross-corr + parabolic sub-sample interpolation, MAE 0.77°
5:20	Features	<i>“Classification. All four feature extractors share the cepstral processing chain: framing, FFT, power spectrum, logarithm, discrete cosine transform.”</i> ■ they differ in the filter bank – power spectrum → bank → log – not in the diagram ■ MFCC mel (speech; validated on ranges) LFCC linear IMFCC inverted mel GTCC gammatone, auditory model, reported more noise-robust ■ grid: 4 extractors $\times$ 3 frame lengths (15/30/50 ms) $\times$ 2 classifiers
6:25	TABLE	<i>“Table one gives the results for the nine-millimetre class.”</i> ■ rows: frame length, four extractors within columns: ACC, RLC, MCC $\times$ SVM, NN ■ MCC: confusion matrix → one value in $[-1, +1]$, <i>not inflated by unbalanced classes</i> ■ GTCC rows highlighted – highest value of all three metrics in every group, both classifiers ■ best: GTCC + NN, 50 ms → 99.15 / 96.77 / 0.98 ■ for scale: 118 test events, 31 of them 9 mm ⇒ 117 of 118 correct, 30 of 31 recalled
7:35	Discussion	<i>“Three observations.”</i> ■ MFCC, LFCC, IMFCC comparable – mel / linear / inverted-mel spacing has little effect ■ GTCC consistently better, every frame length, both classifiers – [CUT] auditory model, known noise-robust: <i>probable</i> explanation, mechanism not isolated here ■ NN slightly ahead of SVM, particularly at longer frames – <i>tendency across the study, not every row</i> (LFCC 30/50 ms goes the other way) ■ longer frames help, <i>but</i> 50 ms increasingly contains the impulse response of the surroundings – <i>“the trend cannot be extrapolated further”</i>
8:45	Conclusion	<i>“To conclude.”</i> ■ distributed network + server → detection, classification, 2D localization ■ cepstral coefficients effective with both classifiers tested ■ GTCC > mel family; NN slightly higher, particularly at longer frames ■ results on real firearms, several calibres and ammunition types
9:30		<i>“Thank you.”</i>

If asked, the numbers. Corpus 371 train / 118 test, five classes (false alarms 86/21, **9 mm 70/31**, 5.56 NATO SD 100/31, 7.62 Tokarev 55/17, .22 60/18) | 26 features per extractor, $\approx 75/25$ split | NN: MATLAB Neural Fitting, Levenberg–Marquardt, 370 iterations | bearing: cross-corr 0.84°, +parabolic 0.77°, PBDE 3.24°, PBDE+linreg 3.91° | detection stage = median-filter trigger, RTSI 2019, down to SNR ≈ 5 dB | appendix slides 11 setup 12 bearing 13 other cepstral 14 metrics (RLC = recall).

Running late? Three passages are marked **[CUT IF LATE]** in the notes – **Motivation** (the three functions), **Signature** (Mach cone), **Discussion** (probable reason GTCC wins). All three ≈ 35 s; dropping them breaks nothing. **Checkpoint**: start the table at 6:25; past 6:55, cut all three.

Do not guess: 2D position error in metres, SNR limits of *this* classifier, firing distances in this dataset, provenance of the waveform on slide 3. *“We did not characterize that in this study – Jakub Svatoš ran the campaign, I can put you in touch.”*